

Exercise 8

Part 1

Solve exercise 13 in Chapter 8 of Ross (p.152).

Part 2

Solve exercise 15 in Chapter 8 of Ross (p.152).

Part 3

Write a program that takes a sample as input and returns the sample median together with a bootstrap estimate of the variance of the median, based on $k = 100$ bootstrap replicates. Then run the program with a sample of $n = 200$ Pareto($\beta = 1, k = 1.05$) distributed observations.

- 1 Compute the sample mean and sample median.
- 2 Estimate the variance of the sample mean using the bootstrap method.
- 3 Estimate the variance of the sample median using the bootstrap method.
- 4 Compare the precision of the two estimators.
- 5 Explain the observed differences in terms of the underlying distribution.